PHYS 225 Winter 2026

<u>Homework #3</u>

1. McIntyre Problem 2.8

2. McIntyre Problem 3.5

3. McIntyre Problem 3.13

4. McIntyre Problem 3.15

5. Complete the following worksheet-style questions about two-state time dependence.

TWO-STATE TIME DEPENDENCE

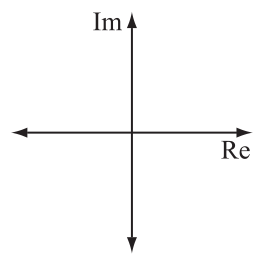
Name ______________________ QM HW-1

Time dependence for stationary and non-stationary states

Suppose that the ***z*-component** of spin for an electron is measured just before $t = 0$ and is found to be $+\hbar/2$. A uniform magnetic field with magnitude B_0 is turned on in the ***z*-direction** at $t = 0$.

1. Write the state vector for this electron as a function of time in the form $\alpha(t)|+\rangle + \beta(t)|-\rangle$ (in the S_z basis). (*Hint:* What are $\alpha(t)$ and $\beta(t)$ at time $t = 0$?)

2. Graph $\alpha(t)$ on the set of axes at right by following the procedure from part I of Worksheet 3.



3. For $t > 0$, what is the probability that a measurement of the ***z*-component** of spin for this electron results in $+\hbar/2$? Justify your answer using your graph.

4. For $t > 0$, what is the probability that a measurement of the ***x*-component** of spin for this electron results in $+\hbar/2$? Show your work. (Assume no measurements were made in the previous question.)

5. Which of the probabilities you computed in questions 3 and 4, if either of them, depend on time?

The term *stationary state* may be used to describe the state of the electron in questions 1 through 5.

6. The state vector for the electron you wrote down in question 1 should be dependent on time. Explain how this is consistent with the fact that the electron is in a stationary state.

7. Consider the student discussion below.

   Student 1: "The probability that a measurement of the z-component of spin resulted in $+\hbar/2$ was independent of time for both the state in questions 1 through 5 of this homework and for the state in part I, section II of Worksheet 3, so they are both stationary states."

   Student 2: "I disagree. The state that started as spin up in the z-direction is stationary because *all* probabilities are independent of time, even though the state vector does depend on time."

   Student 3: "I think it depends on what we measure. Any state is stationary if we measure the spin in the *z*-direction, but not stationary if we were measure the spin in the *x*-direction."

   With which student(s) do you agree, if any? For each student who is incorrect, identify the flaw(s) in that student's reasoning. Explain.

Suppose that the $\boldsymbol{z}$**-component** of spin for a different electron is measured just before $t = 0$ and is found to be $+\hbar/2$ (same as with the first electron). A uniform magnetic field with magnitude B_0 is turned on in the $\boldsymbol{x}$**-direction** at $t = 0$ (different than with the first electron).

8. Express the state vector for this electron as a function of time in the form $\alpha(t)|+\rangle + \beta(t)|-\rangle$ (in the S_z basis). Show your work. (*Hint:* First determine $\alpha(t = 0)$ and $\beta(t = 0)$, then perform an appropriate change of basis to determine the time-dependence before changing back to the S_z basis.)

9. Is the electron described by the state vector in question 8 in a stationary state? Explain.

10. Based on your work in this homework, can an electron be in a stationary state but *not* in an energy eigenstate? Can an electron be in an energy eigenstate but *not* in a stationary state? Explain.